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Creative thinking and academic performance across cultures: Multilevel analyses based on PISA 2022

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Abstract

Background: Creative thinking has been widely recognized as a potential contributor to improved academic performance. However, this association has not been systematically confirmed across culturally diverse societies.

Objective: This study aimed to investigate the relationship between creative thinking and adolescents' academic performance, and to examine the moderating role of two cultural dimensions from the Minkov–Hofstede model: individualism–collectivism and flexibility–monumentalism.

Methods: Using data from the 2022 Programme for International Student Assessment (PISA 2022), multilevel analyses were conducted based on responses from 376,130 adolescents across 48 countries/regions.

Results: The analyses revealed a significant positive relationship between creative thinking and academic performance on a global scale. Furthermore, this relationship was moderated by cultural values: in more individualistic societies, the academic benefits of creative thinking were stronger in reading, math and science; in more flexible societies, creative thinking was more strongly associated with math performance only.

Conclusions: These findings highlight the importance of considering both individual factors and broader sociocultural contexts when examining adolescents' academic performance.

KEYWORDS

academic performance, adolescent, creative thinking, individualism, societal flexibility

Educational relevance and implication

This study underscores the importance of fostering creative thinking to enhance students' academic performance globally. By identifying the moderating effects of cultural dimensions such as individualism and societal flexibility, the findings highlight the need for culturally responsive educational strategies. In individualistic societies, encouraging student autonomy and innovation may further strengthen learning outcomes, while in more flexible societies, integrating creativity into mathematics instruction may be particularly effective. Educators and policymakers should consider both individual cognitive traits and sociocultural contexts when designing interventions to support student achievement.

INTRODUCTION

Creative thinking is widely considered to have significant implications for economic, social and cultural development (OECD, 2024). In today's rapidly evolving knowledge-based society, the ability to think flexibly, divergently and beyond conventional boundaries is valued more than ever (Kleibeuker et al., 2016). Individuals with higher levels of creative thinking are better equipped to respond to new and complex challenges in dynamic environments (de Alencar et al., 2017). Numerous studies have shown that creative thinking plays a critical role in improving students' academic performance, career development and overall well-being (e.g., Acar et al., 2021; Chen et al., 2015; Gajda et al., 2017).

Adolescents, in particular, are at a crucial developmental stage as they transition to greater independence. During this period, they must acquire effective strategies to cope with developmental challenges and conflicts (De Lijster et al., 2018). At the same time, they often face intense academic pressure while adapting to physiological and psychological changes (Zhao et al., 2021). Creative thinking has been identified as a key ability that can support adolescents in problem-solving and conflict resolution (González Moreno & Molero Jurado, 2023), as well as in adapting to their environment and achieving positive outcomes (Anderson & Graham, 2021). Thus, creative thinking is of great value to the academic development of adolescents.

However, most existing studies of the relationship between creative thinking and academic performance have been conducted in single-country contexts, limiting their cross-cultural generalizability. Furthermore, the potential moderating effects of sociocultural factors have often been overlooked. To address these gaps, the present study draws on data from the PISA 2022 dataset, which includes 376,130 adolescents from 48 countries and regions. We aim to examine the relationship between creative thinking and academic performance in a cross-cultural context, and to further explore whether this relationship is moderated by cultural dimensions such as individualism–collectivism (i.e., societal individualism) and flexibility–monumentalism (i.e., societal flexibility).

The relationship between creative thinking and academic performance

Creative thinking refers to the ability of individuals to generate ideas or solutions that are both novel and effective (Beghetto, 2019; Runco & Jaeger, 2012). It is typically characterized by key features such as fluency, originality, flexibility and sensitivity to task constraints (Beghetto, 2019; Runco & Jaeger, 2012). Educational theories have long recognized the strong connection between creative thinking and cognitive processes involved in learning (Gajda et al., 2017). In the revised version of Bloom's Taxonomy, "creating" is placed at the highest level of cognitive skills, highlighting its essential role in the construction of knowledge (Anderson & Krathwohl, 2001). From the perspective of Piaget's

theory of learning, the process of learning involves continuous reconstruction of cognitive structures through assimilation and accommodation—an inherently creative act in itself (Bringuier, 1980).

A growing body of empirical research has examined the relationship between creative thinking and academic performance, yet the findings remain mixed and sometimes contradictory. Broadly speaking, the results can be categorized into two general trends. On the one hand, some studies have identified a positive association between creative thinking and academic performance. For instance, a comprehensive meta-analysis of 120 studies estimated an average correlation of approximately 0.22 between creativity and academic performance, suggesting a moderate but consistent positive link (Gajda et al., 2017). This relationship tends to be stronger when academic performance is measured by standardized test scores rather than GPA (Gajda et al., 2017). In a six-month longitudinal study of 152 middle school students in the United States, Grigorenko et al. (2009) found that creativity-related indicators—such as creative writing tasks and scientific reasoning tasks—could positively predict academic performance. Similarly, Gajda (2016) conducted a study with 1106 Polish students across primary, middle and high school levels, revealing a significant positive correlation between creativity and academic performance, with the strongest effect observed at the middle school level ($r = .29$). Yang and Zhao (2021) have further explored creativity by dividing it into two dimensions—convergent and divergent thinking—and examined how each is differently related to academic outcomes. Specifically, convergent thinking has a stronger positive effect on academic performance than divergent thinking. Several additional studies based on samples from individual countries have also supported the positive link between creativity and academic performance (Freund & Holling, 2008; Mourgues et al., 2016).

On the other hand, some research has reported no significant relationship between creative thinking and academic performance. For example, Atwood and Pretz (2016) found that creativity measures—such as creative self-efficacy and creative achievement questionnaires—did not significantly predict academic outcomes among engineering students. In another study, Gralewski and Karwowski (2012) assessed 589 middle school students from different school types and geographic areas, and concluded that there was no meaningful correlation between creativity and academic performance overall.

Given these conflicting findings, there is a clear need for large-scale, representative data to more accurately estimate the nature of this relationship. Drawing on cross-national data from the PISA assessment, the present study seeks to clarify these inconsistencies by examining the link between creative thinking and academic performance in a global context.

The potential moderating effect of individualism and social flexibility

Previous studies have shown that considering sociocultural context can improve our understanding of academic performance and its predictors. For example, Wang and Li (2024) found that societal individualism moderated the relationship between parental involvement in homework (asking, helping and checking) and elementary school students' academic performance. Specifically, the positive effect of asking behaviours was stronger, while the negative effect of helping behaviours was weaker in collectivistic societies than in individualistic societies. Similarly, Zheng et al. (2024) reported that societal flexibility moderated the link between cognitive flexibility and academic performance, with the academic benefits of cognitive flexibility being more pronounced in more flexible societies. To provide direct evidence on how sociocultural factors may shape the relationship between creative thinking and academic performance, the present study is based on the cultural fit hypothesis and examines the potential moderating roles of two cultural dimensions from the Minkov–Hofstede model: societal individualism and societal flexibility.

The cultural fit hypothesis proposed that when an individual's values, beliefs, or characteristics align with their cultural or societal environment, their performance and well-being will be enhanced (e.g., De Leersnyder et al., 2014; Lou & Li, 2023; Zheng et al., 2024). Consequently, the fit between the core traits of individuals with creative thinking and the characteristics of their sociocultural environment may lead to better academic performance. For example, individualistic societies emphasize independence,

autonomy and personal freedom in thought and action (Welzel, 2013), which encourages independent thinking and challenging traditional ideas (Bukowski & Rudnicki, 2019). These cultural traits align closely with the characteristics needed for creative thinking. Such environments are more tolerant of students' innovative behaviours, enabling them to think independently, question established views, and propose novel ideas in academic activities (Taylor & Wilson, 2012), thus potentially enhancing their academic performance. In contrast, collectivist societies emphasize group cohesion, shared norms and social obligations (Triandis, 2018), often discouraging behaviours that deviate from group expectations (Welzel, 2013). This orientation might conflict with the core features of creative thinking. Consequently, in collectivist cultures, students' creative thinking and behaviours might be constrained due to pressures against deviating from traditional norms (Minkov & Kaasa, 2021), limiting the potential academic benefits associated with creative thinking.

On another dimension, societies with high flexibility promote a self that is adaptable and flexible, adjusting behaviours and values according to changes in the environment and context (Minkov et al., 2018; Minkov & Kaasa, 2022). Human selves are considered flawed, with a significant emphasis on continual improvement and development. This type of culture is open to experimentation and independent exploration (Minkov et al., 2018; Minkov & Kaasa, 2022). As a result, students can respond more flexibly to academic challenges, try innovative solutions and break free from established ways of thinking, thereby improving their academic performance. Conversely, societies characterized by monumentalism emphasize consistency and stability in the self and a high respect and maintenance of fixed norms and values, encouraging positive emotions, including strong self-esteem and pride, which can lead to complacency and hinder self-improvement (Minkov et al., 2018; Minkov & Kaasa, 2022). In such a cultural context, students may tend to adhere to a fixed self-image and follow established conventions. They may overly emphasize self-esteem and self-affirmation, resulting in fewer attempts at innovation or breaking from tradition. This conservative or rigid tendency could weaken the academic benefits associated with creative thinking. Therefore, we propose that in societies with high individualism and flexibility, the relationship between adolescents' creative thinking and academic performance may be stronger.

The present study

Previous studies on the relationship between creative thinking and academic performance have been controversial. Moreover, these studies have several limitations. First, they are predominantly focused on single-country samples, which raises questions about their global applicability. Second, the existing literature lacks an examination of how sociocultural backgrounds influence the relationship between creative thinking and academic performance. Furthermore, while studies have examined the effects of creative thinking on overall academic performance, research on its effects in different subject areas remains inadequate. In response, the present study used data from the 2022 PISA survey to analyse the relationship between creative thinking and academic performance among 376,130 adolescents from 48 countries/regions. In addition, this study examined the potential moderating roles of two cultural dimensions – individualism–collectivism and flexibility–monumentalism – based on the cultural fit hypothesis. Specifically, this study proposed the following hypotheses: There will be a positive correlation between adolescents' creative thinking and academic performance worldwide (Hypothesis 1). In more individualistic societies, there will be a stronger correlation between adolescents' creative thinking and academic performance (Hypothesis 2). In more flexible societies, this correlation may also be stronger (Hypothesis 3).

Additionally, to control for potential confounding effects, this study included age, gender, socioeconomic status, human development index (HDI) and Gini index as control variables at both the individual and country levels. The inclusion of these control variables contributes to identifying the unique effects of sociocultural background in shaping the relationship between creative thinking and academic performance. Through this methodological framework, the study aimed to provide a more

comprehensive perspective on the impact of creative thinking on educational outcomes while exploring how cultural differences affect this relationship.

METHODS

Participants

The individual-level data used in this study were obtained from the Programme for International Student Assessment 2022 (PISA 2022). PISA 2022 assessed approximately 690,000 15-year-old students from 81 countries and economies in mathematics, reading, science and creative thinking. The PISA 2022 was the most recent wave and included all of the key variables examined in this study. A total of 63 countries/regions participated in the creative thinking assessment. However, due to a lack of data on country-level variables in some countries/regions, the final sample for data analysis included 376,130 students from 48 countries. Table 1 shows the number of students and the number of countries/regions for each variable. Bryan and Jenkins (2016) suggest that a sample of at least 25 countries/regions is necessary to ensure the robustness of country-level effects; thus, the current study's sample size has sufficient statistical power.

It is worth noting that an important prerequisite for cross-national comparisons is the establishment of measurement invariance (Davidov et al., 2018). In PISA 2022, methods such as expert reviews, rigorous translation and adaptation processes, and statistical analyses using multi-group item response theory (IRT) models were employed to ensure measurement invariance across countries and languages (OECD, 2022a, 2022b). These approaches ensured the global consistency of the assessment tools and the international comparability of the results, thus ensuring the validity and reliability of the PISA data (OECD, 2022a, 2022b).

Creative thinking

The PISA 2022 creative thinking assessment was structured around three core dimensions: generate diverse ideas, generate creative ideas, and evaluate and improve ideas. These dimensions included 12, 11 and 9 tasks respectively, distributed across four domains: written expression, visual expression, social

TABLE 1 Descriptive statistics.

	<i>N</i>	<i>M</i>	<i>SD</i>	Min	Max
Individual-level variables					
Creative thinking	499,843	28.45	12.01	0.15	58.33
Reading	499,843	437.17	103.92	17.88	827.48
Math	499,843	441.67	97.75	128.37	841.87
Science	499,843	450.50	99.88	79.66	858.74
Gender	499,770	/	/	1 = female	2 = male
Age	499,843	15.79	0.29	15.17	16.42
ESCS	479,645	−0.34	1.14	−6.84	7.38
Country-level variables					
Individualism–Collectivism	52	−6.40	81.62	−171	182
Flexibility–Monumentalism	52	3.31	88.43	−187	199
HDI	61	0.84	0.08	0.67	0.96
Gini Index	59	34.45	6.75	22.10	48.50

Abbreviations: ESCS, Index of economic, social and cultural status; HDI, Human Development Index.

problem-solving and scientific problem-solving. Responses were evaluated by trained human raters using detailed scoring guidelines outlined in coding manuals, considering factors such as originality, relevance and elaboration. Each response was assigned either full credit, partial credit, or no credit. To ensure fairness, interpretability and international comparability, PISA 2022 employed IRT to estimate students' latent ability levels (θ) based on their final scores. Subsequently, through a nonlinear transformation using the test characteristic curve (TCC), students' θ values were converted into estimates of the number of items they could potentially answer correctly (number correct), resulting in 10 plausible values for each student's creative thinking ability (OECD, 2022a, 2022b).

Academic performance

Students' plausible values in reading, mathematics and science were used as indicators of academic performance. In PISA 2022, each student received 10 plausible values for reading, mathematics and science, respectively, representing their potential levels of ability in these domains. The plausible values refer to a range of scores that students might achieve based on their responses to test questions and additional background information, providing estimates that reflect the likely range of their ability (OECD, 2022a, 2022b).

Cultural values

National scores for cultural values were obtained from the Minkov–Hofstede cultural dimension model (Minkov & Kaasa, 2022). The Minkov–Hofstede model represents an extension and update of Hofstede's original cultural dimensions (Minkov & Kaasa, 2022). This model includes two cultural dimensions: individualism–collectivism and flexibility–monumentalism, providing an objective framework for understanding and analysing differences in social behaviours across countries and regions (Minkov et al., 2018; Minkov & Kaasa, 2022). Higher scores indicate higher levels of individualism and societal flexibility within a country/region.

Individual-level and country-level covariates

At the individual level, we included adolescents' age, gender (1 = female; 2 = male) and the index of economic, social and cultural status (ESCS), representing their socioeconomic background, as covariates.

At the country level, we used the HDI and the Gini index as covariates, as both have been identified as strong predictors of academic achievement (e.g., Eriksson et al., 2020; Workman, 2023). Specifically, HDI reflects national-level development by measuring average achievement across key dimensions, including life expectancy, education level and gross national income per capita, with scores ranging from 0 to 1 (higher scores indicate higher levels of development). HDI data were obtained from the United Nations Development Programme (<https://hdr.undp.org/data-center/documentation-and-downloads>). The Gini index measures national-level inequality, indicating the extent to which income or consumption distribution among individuals or households deviates from perfect equality. Scores range from 0 (perfect equality) to 100 (maximum inequality), with higher values indicating greater inequality. Data on the Gini index were sourced from the Standardized World Income Inequality Database (SWIID; Solt, 2020).

Statistical analysis

Given that student data were nested within different countries or regions, we conducted a series of multilevel models, using students' subject-specific scores as the outcome variables. First, we ran unconditional (empty) models – without any predictors – to calculate the intraclass correlation coefficient (ICC) for each academic subject. ICC values reflect the degree of between-country variation in academic performance, with Cohen (1988) recommending multilevel modelling when ICC values exceed 0.059. Next, adolescents' creative thinking was entered as an individual-level predictor. At the national level, cultural values were included both as predictors of academic performance intercepts (indicating main effects of cultural values) and as predictors of the slopes of creative thinking (representing cross-level interactions), while controlling for other covariates. All continuous individual-level variables were group-mean centred within each country/region, whereas all country-level variables were grand-mean centred.

RESULTS

Table 1 shows the descriptive statistics for the variables in this study. Country-specific descriptive statistics can be found in the [Supporting Information](#).

The ICC was 0.2626 for reading, 0.3168 for math and 0.2782 for science, indicating that variances between countries accounted for approximately 26.26% of the total variance in reading, 31.68% in math and 27.82% in science. These results also confirmed the necessity of the application of multilevel modelling. Table 2 shows the results of the multilevel analyses. At the individual level, adolescents' creative thinking was a strong positive predictor of their performance in all three academic domains – reading, math and science. At the country level, the HDI positively predicted academic performance in all three domains, whereas social flexibility was a positive predictor only for math and science.

More importantly, we identified significant cross-level interactions. Specifically, the interactions between individualism and creative thinking were significant in predicting academic performance across all three subjects. This indicates that in more individualistic societies, the positive association between adolescents' creative thinking and academic performance is stronger. In contrast, the interaction between societal flexibility and creative thinking was significant only for math, suggesting that in more flexible societies, creative thinking contributes more to students' performance in math. However, no significant interaction was found for reading or science, suggesting that the relationship between creative thinking and performance in these two subjects does not vary significantly by level of societal flexibility.

DISCUSSION

In this study, we comprehensively examined the relationship between adolescents' creative thinking and academic performance across cultural contexts and investigated whether this relationship is moderated by individualism–collectivism and flexibility–monumentalism. Our findings have important implications for the development of flexible teaching strategies and culturally adaptive educational policies aimed at enhancing students' creative thinking and academic performance.

Our findings supported Hypothesis 1 and were consistent with previous research indicating that creative thinking is positively associated with better academic performance (Freund & Holling, 2008; Gajda et al., 2017; Grigorenko et al., 2009; Mourgues et al., 2016). This study further highlighted the significant positive role of creative thinking during adolescence. Previous studies have shown that the association between creative thinking and academic performance is particularly strong during adolescence, especially in secondary school (Gajda, 2016; Gajda et al., 2017). Our findings, based on adolescents from 48 countries, provide robust evidence to support and extend these previous conclusions. Adolescence is

TABLE 2 Estimated results of the multilevel model (with robust standard errors).

	Reading (model 1)	Math (model 2)	Science (model 3)
	Coeff (SE)	Coeff (SE)	Coeff (SE)
Level 1 – individual level			
Intercept (γ_{00})	0.015 (0.039)	0.027 (0.040)	0.028 (0.039)
Creative Thinking (CT; γ_{10})	0.801*** (0.012)	0.780*** (0.013)	0.799*** (0.011)
Gender (β_2)	−0.004 (0.004)	0.000 (0.002)	−0.002 (0.002)
Age (β_3)	−0.000 (0.001)	−0.001 (0.001)	−0.000 (0.001)
ESCS (β_4)	0.001 (0.001)	0.000 (0.001)	0.001 (0.001)
Level 2 – country level			
Individualism (γ_{01})	0.071 (0.080)	0.033 (0.066)	0.077
Flexibility (γ_{02})	0.048 (0.054)	0.214*** (0.055)	0.133* (0.050)
HDI (γ_{03})	0.345*** (0.084)	0.264*** (0.060)	0.273*** (0.066)
Gini Index (γ_{04})	0.025 (0.047)	−0.025 (0.040)	0.010 (0.040)
Cross-level interactions			
Creative thinking × Individualism (γ_{11})	0.087*** (0.012)	0.065*** (0.015)	0.092*** (0.013)
Creative thinking × Flexibility (γ_{21})	0.004 (0.013)	0.056*** (0.014)	0.020 (0.013)
Residual variance			
Variance Component (Standard Deviation)			
Level 1 variance (r_{ij})	0.292 (0.541)	0.308 (0.555)	0.307 (0.554)
Level 2 variance of intercept (u_{0j})	0.075*** (0.274)	0.070*** (0.264)	0.070*** (0.264)
Level 2 variance of slope of CT (u_{1j})	0.006*** (0.079)	0.007*** (0.085)	0.005*** (0.073)
Number of individuals	376,130	376,130	376,130
Number of countries/regions	48	48	48
Goodness-of-fit (Deviance)	605285.95	625252.56	624144.95

Note: * $p < .05$; ** $p < .01$; *** $p < .001$.
Abbreviations: ESCS, Index of economic, social and cultural status; HDI, Human Development Index;

a unique developmental phase characterized by contradictions and rapid change (De Lijster et al., 2018; Smetana et al., 2005), during which individuals increasingly question authority and explore their identities, creating favourable conditions for the development of creative thinking (Gajda, 2016). During this stage, adolescents not only have to cope with significant physical and psychological changes, but also with significant academic pressures (Zhao et al., 2021). Creative thinking can strengthen adolescents' problem-solving and decision-making skills, thereby improving their cognitive abilities and academic task completion (Sun & Hui, 2012). Importantly, adolescents who demonstrate stronger creative thinking often show greater academic resilience and better cope with academic challenges (Qi, 2025). Therefore, promoting creative thinking may help adolescents adapt to changing educational environments and enhance their academic performance.

Moreover, this study found meaningful cross-level interactions between cultural dimensions (individualism and social flexibility) and adolescents' creative thinking. Specifically, the association between creative thinking and academic performance was stronger in individualistic societies, supporting Hypothesis 2. Individualistic cultures emphasize independence and self-expression (Welzel, 2013). Such cultures encourage students to explore independently, challenge assumptions and generate innovative ideas—qualities that are directly related to creative thinking, thus enhancing the academic benefits of creative thinking. In contrast, collectivistic cultures emphasize group cohesion and adherence to shared

norms (Welzel, 2013), values that are less aligned with the demands of creative thinking, such as challenging convention or questioning authority. Thus, students in collectivistic societies may limit their creative expression, avoid challenging teachers or peers to maintain group harmony, or over-rely on authority for fear of violating established norms (Ping et al., 2024), potentially diminishing the academic benefits of creative thinking.

Notably, our results indicated that social flexibility only moderated the relationship between creative thinking and math performance, but not reading or science performance. This finding partially supported Hypothesis 3. We propose two possible explanations for this finding. First, it may reflect the strong influence of East Asian Confucian cultures on math performance (Leung, 2001; Zheng et al., 2024). Societal flexibility, a dimension largely derived from Confucian cultural traits prevalent in East Asia, has consistently valued math as a central discipline and has shown significant advantages in math performance (Leung, 2001; Minkov et al., 2018). Minkov and Kaasa (2022) also demonstrated a strong positive correlation (approximately 0.8) between societal flexibility and national math scores. Thus, the unique impact of societal flexibility on mathematics achievement may stem from its roots in East Asian Confucian cultures, which have traditionally emphasized math education.

In addition, different academic subjects may require different degrees of creative thinking. Societal flexibility emphasizes adaptability, delayed gratification and personal development (Minkov et al., 2018), characteristics that are particularly beneficial for math learning. Prior research has shown that math performance is closely linked to students' sustained effort and ability to delay gratification (Herndon et al., 2015; Yu et al., 2021). These traits are strongly related to social flexibility and facilitate students' ability to maintain motivation even when the immediate outcomes of math learning are unclear (Stupel & Ben-Chaim, 2017). Furthermore, math problems often have multiple valid solutions (Stupel & Ben-Chaim, 2017), allowing for adaptive thinking and flexible problem-solving approaches favoured by cultures high in societal flexibility. In contrast, science education generally emphasizes structured mastery of basic concepts, factual knowledge, theories and systematic experimental skills (National Research Council et al., 2012; OECD, 2022a, 2022b). Reading focuses primarily on integrated language skills, cultural understanding and contextual strategies (OECD, 2018). These disciplines typically require standardized skill application and factual knowledge acquisition, which may be less aligned with the flexibility and autonomy promoted by social flexibility cultures.

PRACTICAL IMPLICATIONS

The current findings revealed two important practical implications. First, creative thinking interventions can be considered an effective means of improving academic performance. Educators should explore how resource allocation and environmental optimization can support the development of students' creative thinking. For example, “student-centered learning” and “project-based teaching” for solving complex real-world problems have been shown to be effective teaching strategies for enhancing creative thinking and deepening learning (Albar & Southcott, 2021; Horng et al., 2005). Second, when designing educational interventions, it is critical to consider the sociocultural backgrounds. For instance, in an individualistic society, enhancing training in creative thinking may more effectively and directly improve academic performance. Conversely, in collectivistic cultural settings, fostering creative thinking alone may have relatively modest benefits for academic performance. However, adopting strategies that are consistent with collectivist cultural values, such as collaborative group work, can foster students' collective intelligence, collaborative skills and academic performance. This approach not only enhances students' social interaction skills but also helps develop broader cognitive and problem-solving skills.

LIMITATIONS AND FUTURE DIRECTIONS

Despite providing valuable insights, this study has several limitations that should be noted. First, although the multinational assessment data used in this study provide strong representativeness (Pitsia, 2022), its cross-sectional nature limits the ability to make causal conclusions (Rutkowski & Delandshere, 2016). Therefore, future research could use longitudinal or experimental designs to explore the long-term influence of creative thinking on academic performance and to establish clearer causal relationships across different educational stages. Second, data were collected primarily through self-report measures, which may introduce response bias due to respondents intentionally or unintentionally providing inaccurate answers. Future studies could use more objective measures, such as third-party assessments or implicit testing approaches, to improve measurement validity. Third, our study focused primarily on academic performance in math, reading and science based on PISA data. Future research could expand the scope by including other subjects, such as art and design, as these areas may have a more direct and stronger connection to creative thinking (Dou et al., 2021). Finally, this study only examined the moderating effects of sociocultural factors on the relationship between creative thinking and academic performance. Future studies could also consider the role of individual factors such as self-efficacy, personality and motivation, as these variables may significantly influence the relationship between creative thinking and academic performance (Fan & Sun, 2024).

CONCLUSION

In the era of globalization and the knowledge economy, improving adolescents' academic performance has become a key concern for educational research and policymaking worldwide (Wiseman, 2010). The current findings reveal a significant positive relationship between creative thinking and adolescents' academic performance in an international context. More importantly, in societies characterized by higher individualism, adolescents' creative thinking is more strongly linked to their academic performance in reading, mathematics and science. In contrast, in societies with higher levels of societal flexibility, creative thinking is associated more strongly only with math performance. These results emphasize that to fully understand the relationship between creative thinking and academic performance, it is necessary to consider not only individual factors but also the role of sociocultural contexts.

AUTHOR CONTRIBUTIONS

Yueling Wei: Writing-original draft; writing-review and editing. Wang Zheng: Conceptualization; methodology; writing-original draft; writing-review and editing; supervision. Yinqiu Zhao: supervision; writing-review and editing. Zhiyu Liu: writing-review and editing; supervision.

FUNDING INFORMATION

None.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data for this study were obtained from public databases, and the source of the data has been reported in the text.

ETHICS STATEMENT

The data used in this study were obtained from publicly available databases and had no ethical implications.

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How to cite this article: Wei, Y., Zheng, W., Zhao, Y., & Liu, Z. (2025). Creative thinking and academic performance across cultures: Multilevel analyses based on PISA 2022. *British Journal of Educational Psychology*, 00, 1–13. <https://doi.org/10.1111/bjep.70045>